



Overview of Oil and Gas Sector

Feb 2024

Table of Contents

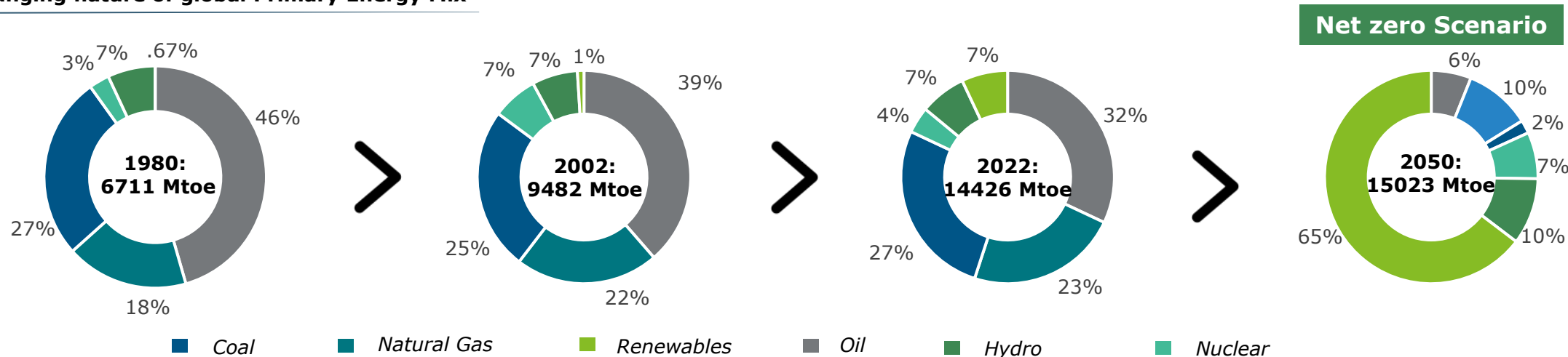
-
- 1 Global Energy Mix | Overview
 - 2 Global Oil & Gas | Production & Consumption
 - 3 Global Oil & Gas | Emerging Trends
 - 4 ESG / Sustainability and Digital initiatives
 - 5 India Energy Mix | Overview
 - 6 India Oil & Gas | Production & Consumption
 - 7 India Oil and Gas sector | Reforms and Initiatives
 - 8 India Oil and Gas sector | Natural gas sector
 - 9 Challenges and Enablers for Indian Oil and Gas Sector
-



1 Global Energy Mix | Overview

The global energy system is undergoing transition, but **Oil & Gas are still the mainstay of global energy and economic growth accounting** for over half of the global consumption.

Changing nature of global Primary Energy Mix



Coal: Coal share is stagnant with sharp drop in developed countries and has been replaced primarily by **gas and renewables**, however, a large pipeline of projects exists in India and China.



Oil: While still dominant in transportation, oil demand is **projected to plateau soon** due to electric vehicle adoption and fuel efficiency improvements.



Natural Gas: Consumption continues to rise, particularly in regions replacing coal-fired power plants emerging as a bridge fuel. However, its future growth faces competition **from renewables**.



Nuclear: After initial setbacks, nuclear is experiencing a cautious revival driven by its **carbon-free nature** and reliability.



Hydro: Growth has slowed due to **environmental concerns** and limitations on building new dams. However, it remains a crucial source of clean energy, especially in developing countries.



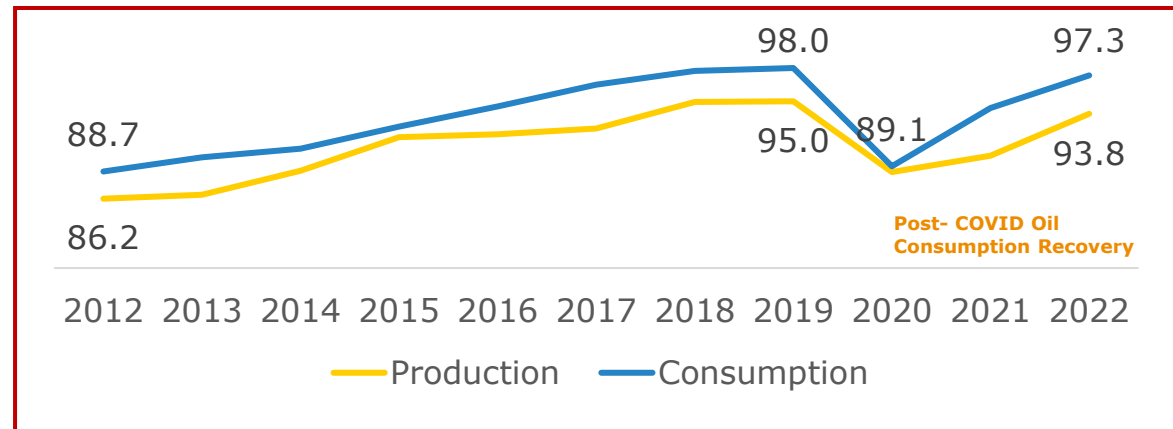
Renewables: Solar and wind power are undergoing remarkable growth, driven by falling costs and technological advancements. They are expected to be the leading sources of future energy demand, particularly in electricity generation.

Sources: 1. Statistical Review of World Energy Data, 2023, , BP Stats 2023, Internal analysis

2 Global Oil & Gas | Production & Consumption

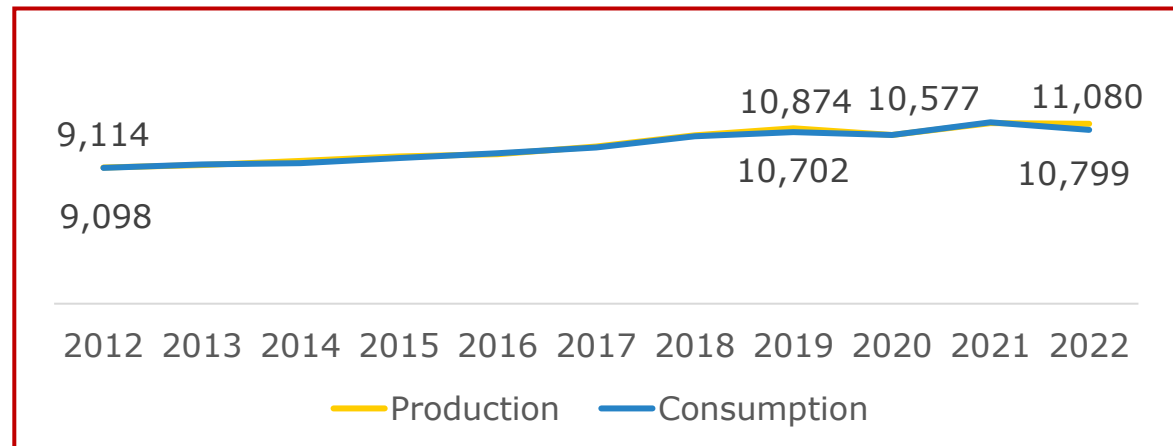
The production and consumption are likely to witness a positive trajectory over the coming years; while China & India would be the strongest consumer, the US and LAMEA (Latin America, the Middle East and Africa) are expected to be the strongest producers.

Global crude oil production and consumption, million barrels per day (mb/d)



- The year 2020 witnessed a decrease due to the economic and mobility impacts of the pandemic, including widespread lockdowns globally but the demand has now rebounded
- Consistent global oil inventory capacity building expected over the next few years as global oil production is likely to be strong with supply growth led by **US, Canada, Brazil**.
- Global crude oil consumption is likely to increase at a CAGR of ~2% over 2022-28, driven by demand from the developing nations like **India, China and the Middle East**.

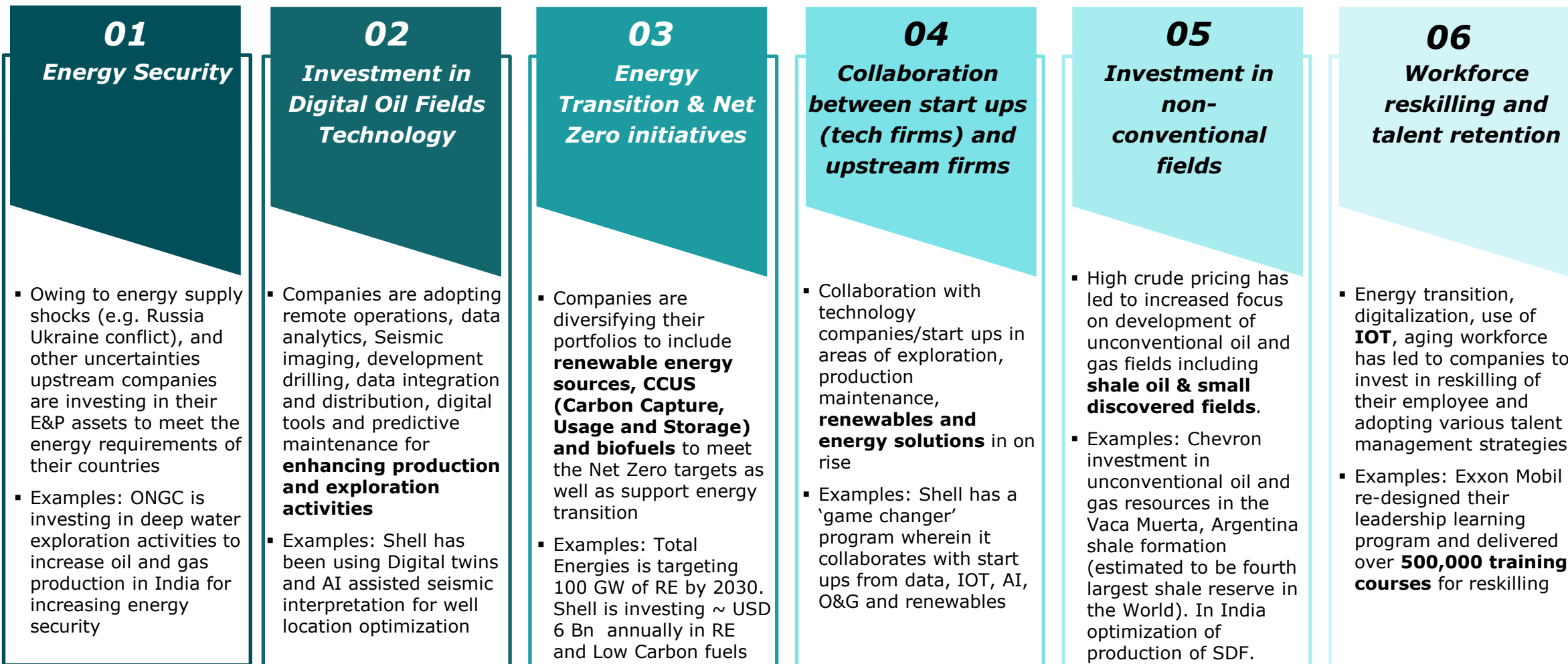
Global natural gas production and consumption, billion cubic meters



- Global natural gas consumption rebounded by in 2021 YoY, after the decline seen in 2020 due to Covid pandemic
- The natural gas consumption slowed down over 2022, as Russia-Ukraine war has pushed up prices and fueled supply disruption fears
- APAC (Asia Pacific) will drive the incremental global gas consumption in the coming years, driven principally by the development of gas in **India and China**.

3 Global Oil & Gas | Emerging Trends - Upstream

The upstream oil and gas sector is embracing digital tech, diversifying with renewables, partnering with tech firms, developing unconventional resources, and reskilling its workforce for energy transition



3 Global Oil & Gas | Emerging Trends - Midstream

The midstream sector is witnessing growth in LNG capacity, long-term gas contracts, pipeline investments, investments in innovative gas delivery systems, hydrogen blending for decarbonization, etc

01

Revival of LNG projects

- The increasing demand for NG has led to multiple producers reviving Liquefaction trains and customers ordering new LNG vessels to secure suppliers.
- Example: ~188 Mn MT liquefaction capacity has FID / under construction. Further, 1,036 Mn MT capacity is announced/pre FID. Total 1100+ Liquefaction terminal with 4% CAGR.

02

Increase in long term contracts for LNG

- Long term LNG contracts are signed by importers to **secure energy supplies** for their countries. The recent supply shocks in LNG market due to Ukraine – Russia conflict has also contributed to this trend
- Example: In 2022, a record 81 MTPA of long-term LNG contracts were signed with majority of it being with US based suppliers

03

Augmenting midstream infrastructure to improve connectivity

- Midstream companies are making investments in pipelines, storage facilities (strategic reserves), liquefaction and re-gas terminals to support energy security and energy transition.
- Example: In terms of trunk pipelines length, India leads among countries with an upcoming length of 29,881 km from 2023-2027

04

Innovation in delivery of LNG to customers – virtual pipelines (CNG/LNG)

- To increase penetration of natural gas, companies are developing new delivery systems such as **CNG cascade and LNG** to reach areas without pipeline connectivity.
- Example: **Exxon Mobil** India LNG and **IOCL** have invested in virtual pipeline to expand gas access in India

05

Hydrogen blending in natural gas

- Companies are investing in R&D for **hydrogen blending for decarbonizing** hard-to-abate sectors, including options of peaking and load-following electricity and industrial heating
- Example: Utility companies such as GAIL and British Gas are undertaking pilot projects for hydrogen blending in natural gas pipeline network

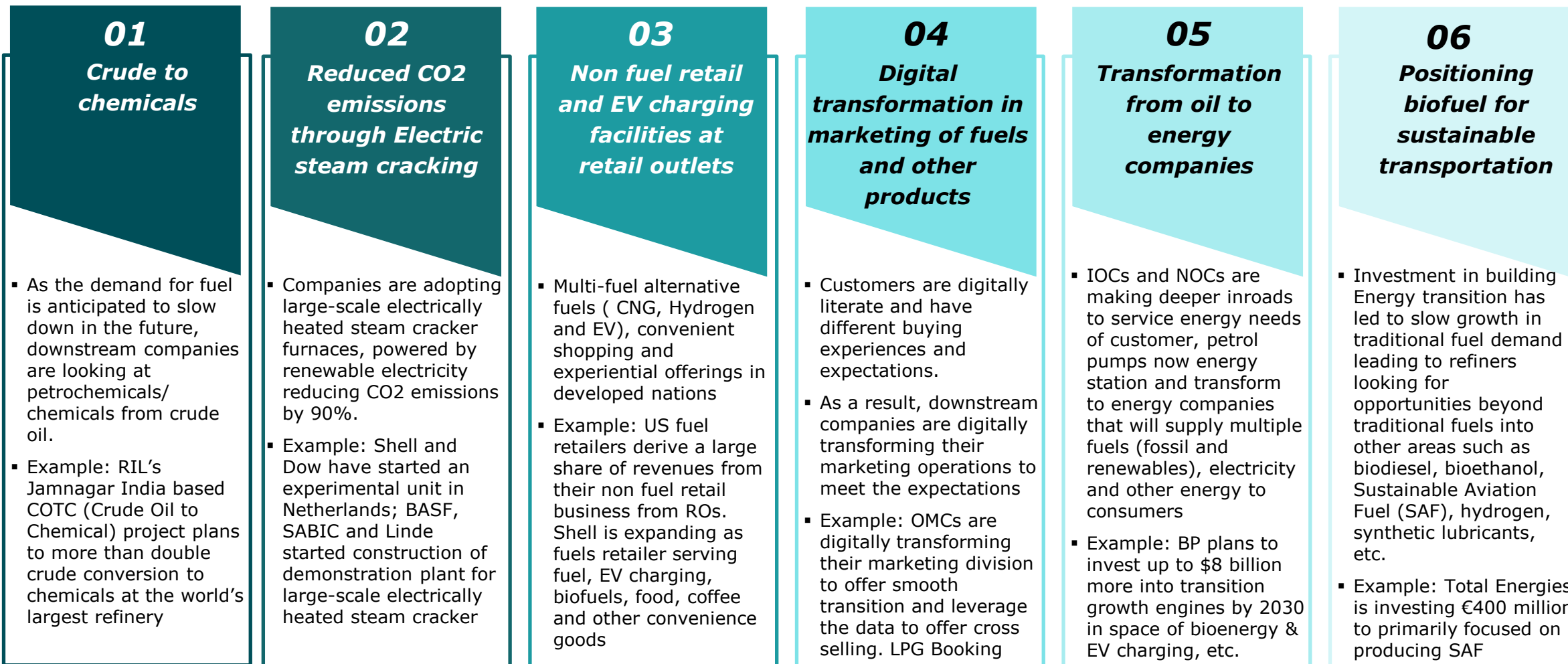
06

Investments in strategic petroleum reserves for energy security

- Investment in building strategic petroleum reserves to absorb supply shocks. Diversifying supply chain made up of pipelines, storage, export terminals to process fossil and renewables are key trends in midstream sector
- Example: Strategic gas reserves have been planned in Europe. Similarly, ISPRL in India is planning to increase SPR capacities

3 Global Oil & Gas | Emerging Trends - Downstream

Downstream sector is witnessing an increase in shift to petrochemicals, focus on sustainable technologies, digital transformation, diversified energy services, and offering alternative fuels and non fuel products/services

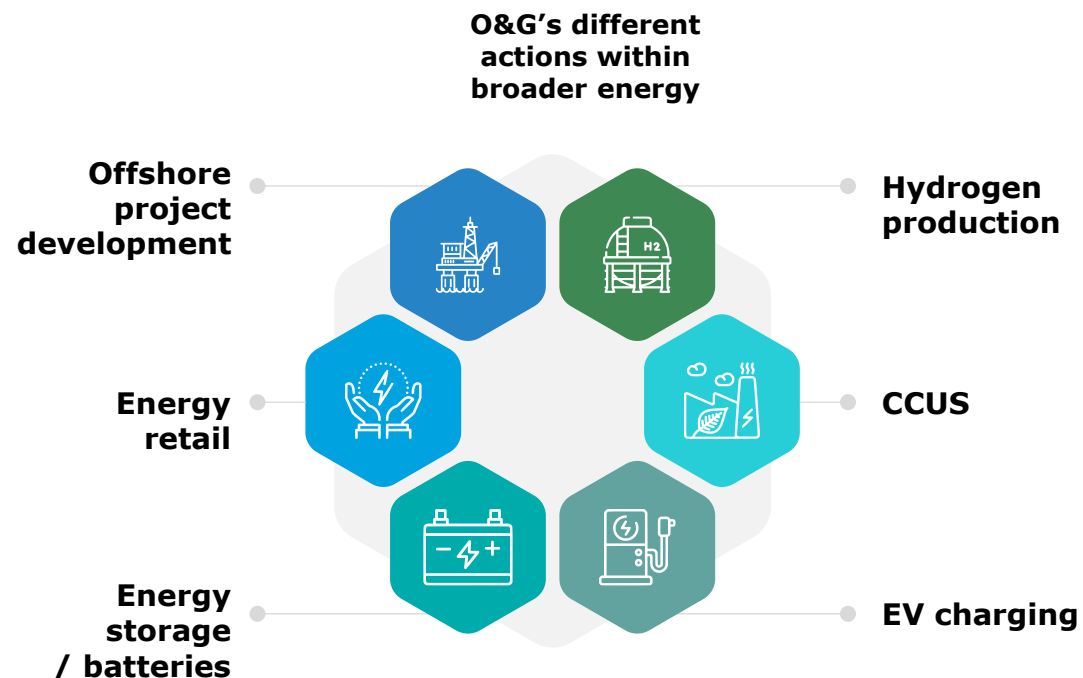


3 Changing Business Areas & Models| Shift from traditional O&G business to energy

Oil & gas companies are diversifying as energy providers, more so with decarbonization targets – via investments in alternative low-carbon fuels or renewables, like green hydrogen and natural gas, which is less damaging to the environment

Overview:

- **Large Oil & Gas players are switching to energy business.**
- **Transition from oil & gas to energy might not be very comfortable but is surely a way forward to manage transition risks.**
- **Companies need ensure supply security and excel amidst stringent carbon-reduction targets.**
- **Uncertainty looms around support activities such as offshore generation, EV charging, and hydrogen production.**
- **New and legacy businesses are briskly changing their business models, or adapting to the change.**
- **Players are resorting to different ways – organically developing a new product; or partnering with established energy players.**



4 Energy transition – ESG / Sustainability initiatives

Oil and gas companies are facing intense pressure from investors, regulators and activists to cut down on their emissions, and they are now introducing aggressive climate targets

Drivers of climate and sustainability in Oil and Gas Sector



REGULATORY PRESSURE

O&G companies continue to tackle new rules from regulatory bodies to cut down on their emissions



INVESTORS

Companies are facing increasing pressure from investors to adopt emission reduction and decarbonization strategies



ACTIVISTS

Mounting pressure from activists globally is forcing O&G companies to adopt sustainability initiatives

Initiatives taken by select O&G giants



- Shell announced plans to produce 2.2 million tons of **sustainable aviation fuel (SAF)** by 2025, and for SAF to amount to at least 10% of all the aviation fuel it produces by 2030.
- Has deployed **digital twins in Norway** for optimizing production and energy use; this helps collect real-time data around carbon emissions and energy usage
- Investing in **digital hydrogen plant, leveraging electrolyzers**
- Launched **used oil management service** in IND



- The company is taking action to **reduce its Scope 1 & 2 emissions** from its operated industrial facilities **by ~40% by 2030** – via improving energy efficiency, reducing flaring, reducing methane emissions, supplying renewable electricity, and using CCS for residual emissions; new target of lowering **Scope 3 emissions from petroleum products sold globally by 30%+ between 2015 and 2030**

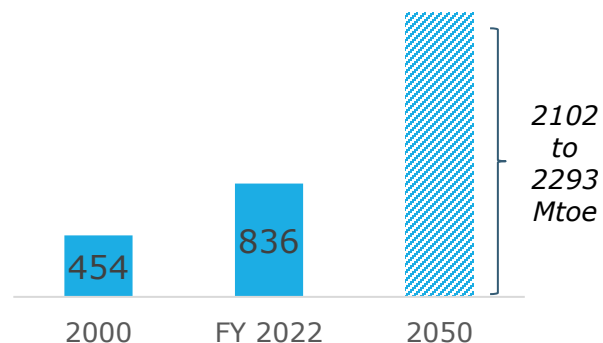


- Pledged to invest **USD\$10.1 billion** (INR750 billion) to increase the production capacity from **renewable energy sources** such as solar and hydrogen
- As a part of its recently announced sustainability targets, it plans to achieve carbon neutrality by 2035. It is also investing in solutions such as **carbon recycling and carbon capture** and storage technologies

5 Overview of India Energy Mix

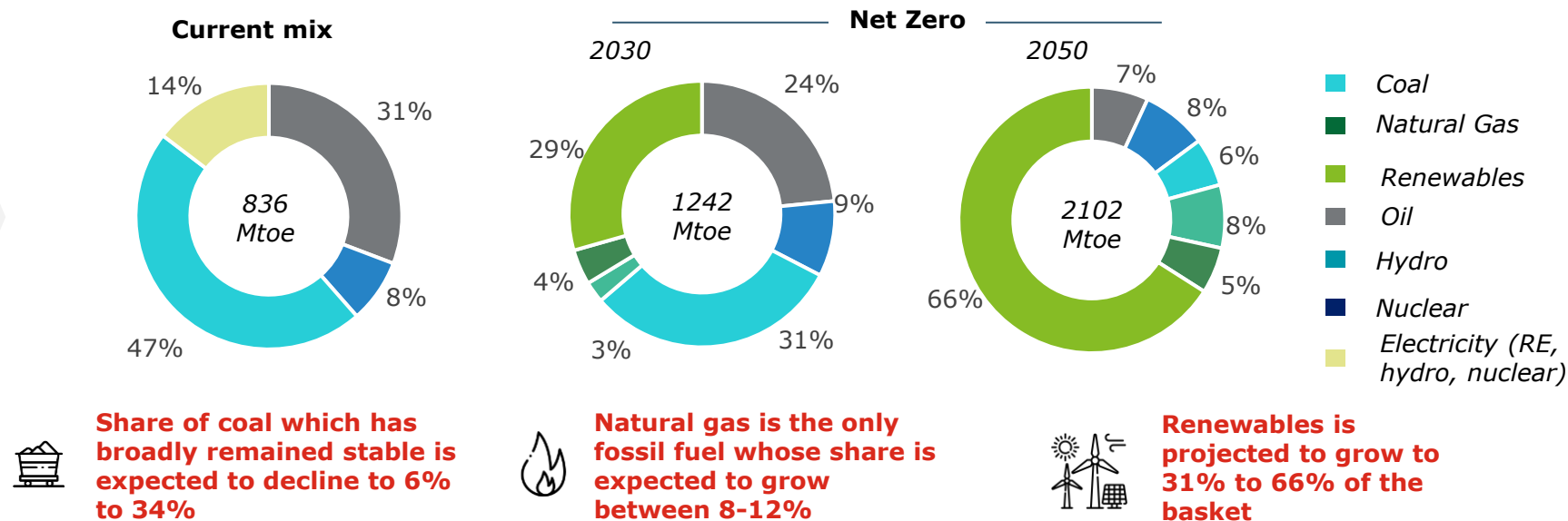
India leads the growth in primary energy which at present is predominately met by fossil energy sources with strong pressure to transition to low or zero emission sources for meeting future and existing needs

Primary Energy Consumption (Mtoe)



- In every scenario, energy consumption is projected to grow more than 2x
- Growth is underpinned by increasing population, industrialization and prosperity

Various scenarios have projected the energy mix in 2050 depending on the path adopted to meet increased energy needs



Trends (Growth)

Robust economic growth @ 6-7% will propel the consumption of energy in the coming decades

Renewed focus on increasing share of manufacturing in the GDP

Increasing population is expected to add ~25 crore people by 2050

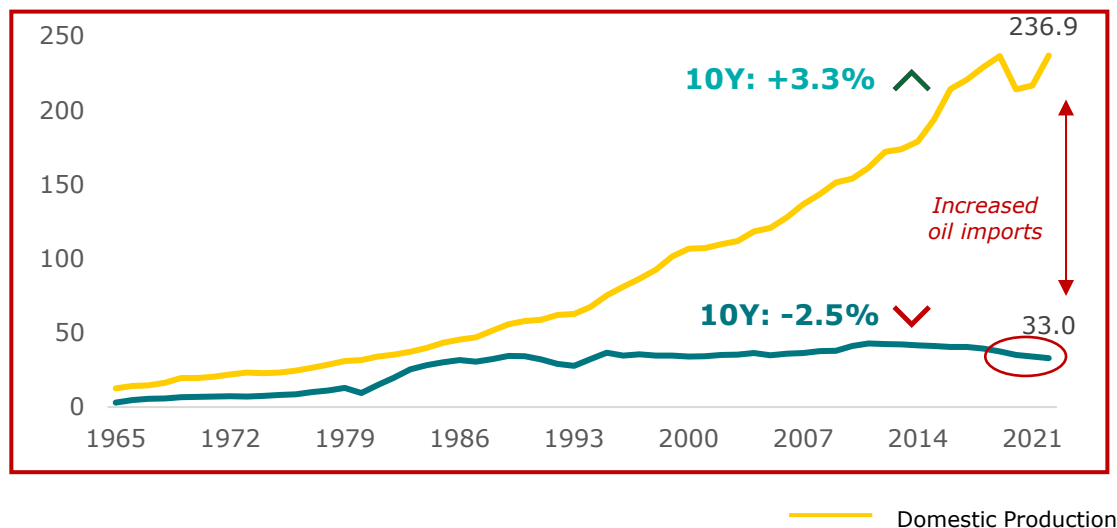
Urbanization and built environment will increase the need for steel, cement, appliances and cooling

Mobility
Increasing ownership of private vehicles and focus on building of mass mobility and infrastructure projects.

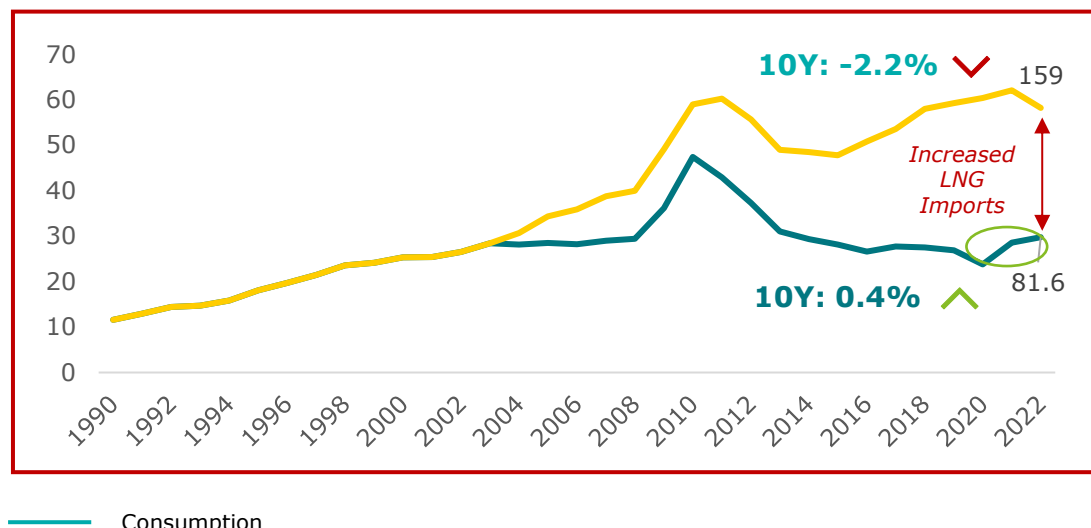
6 India Oil & Gas | Production & Consumption

With increasing energy needs over next several decades, India need to reverse the decline and increase domestic production of oil and natural gas

Oil Production and Consumption (MMT)



NG Consumption and Production (MMSCMD)



Threats and Concerns

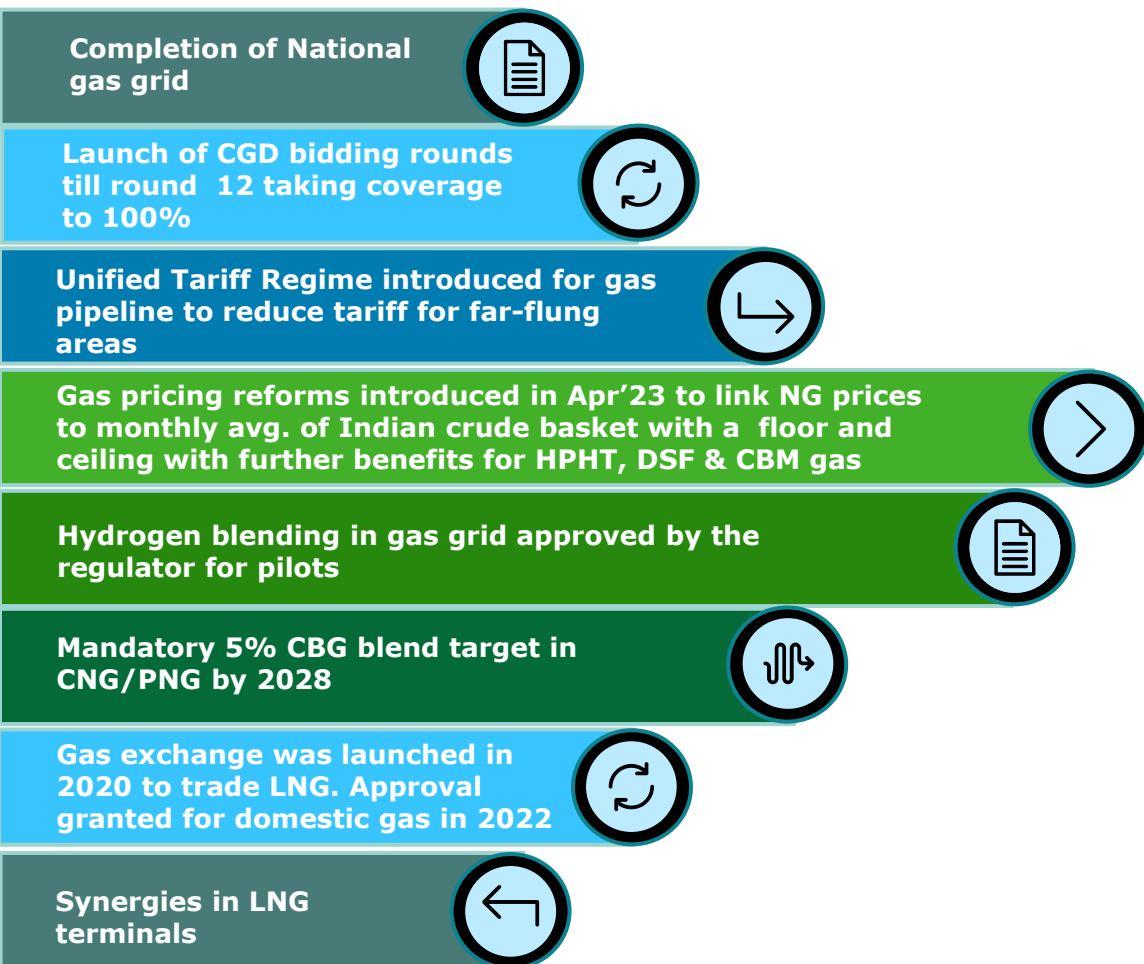
- Oil consumption in India in last 10Y has grown at a CAGR of 3.3% while the production has dropped at a CAGR of -2.5% resulting in increased reliance on imports which has climbed to >86%
- Further, most of the producing fields are mature and there have been no major discoveries over the last decade barring the KG basin which also has been delayed.
- There is a dire need to reverse the decline by increasing production from existing fields and bringing online new discoveries at the earliest

- NG consumption has stagnated in the last 10Y due to unavailability of sufficient cheap domestic gas and expensive and volatile LNG
- Domestic production has continuously dropped since the peak reached in 2010 from incremental gas from KG D6 fields of RIL. This has concurrently led to rise in LNG imports to plug the gap
- The decline has reversed in the last 2Y and is expected to grow with major contribution from KG basin fields of RIL (MJ, R-cluster, satellite) and KG DWN 98/2 of ONGC

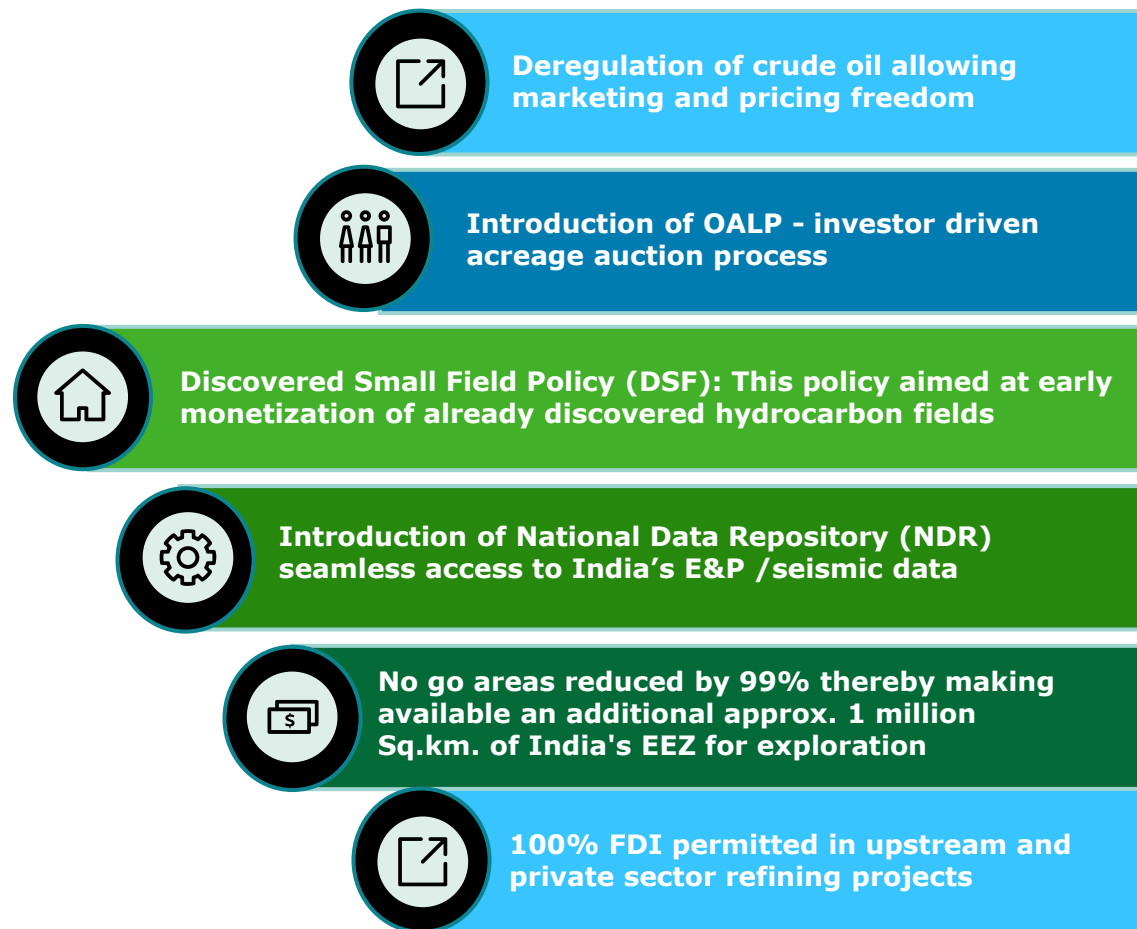
7 India Oil and Gas sector | Reforms and Initiatives

Government and PNGRB is enabling the growth of oil and gas sector in India through various reforms and initiatives

Natural Gas



Crude Oil



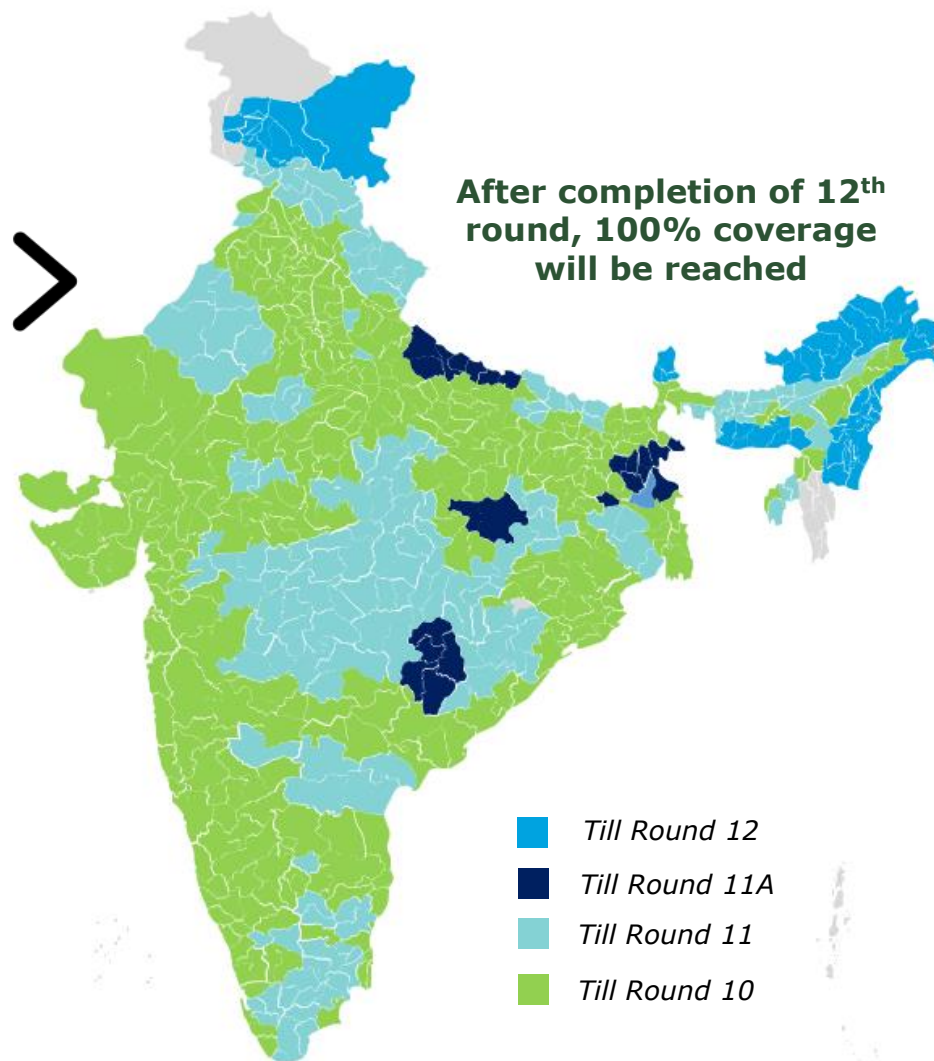
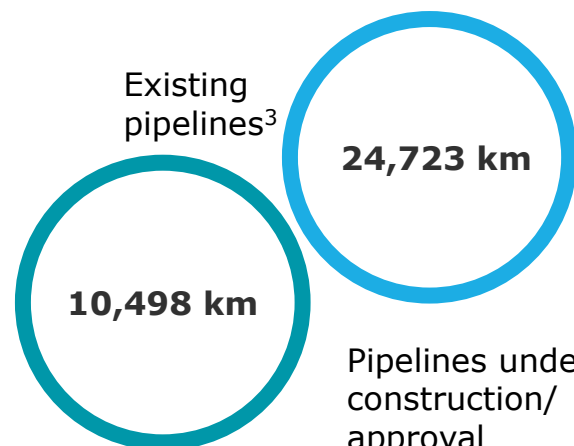
8 India Oil and Gas sector | Natural Gas sector

The coverage of natural gas has grown across the length and breadth of the country under the stewardship of PNGRB

City Gas Distribution: Till 12 bidding round, coverage of CGD will expand to

- 1 **308 GAs**
- 2 **28 states and UTs, 640+ districts**
- 3 **99% of area covering 99% population**

National Gas Grid



Domestic PNG connections¹

1.22 crore

Industrial & Commercial PNG connections²

58.3 thousand

CNG Stations¹

6,258

Notes: 1. 1. As of Dec 2023, 2. As of April 2023, 3. Includes partially commissioned, 4. Map not to scale

Sources: 1. PNGRB, PPAC, MoPNG, Press articles

9 Challenges and Enablers for Indian Oil and Gas Sector

Further enablers to overcome existing challenges in oil and gas sector

1

Increasing exploration and high-quality data to improve perception of perspective to attract global Oil and gas majors.

2

Developing mature trading hubs for rational and competitive price discovery of Natural gas

3

Inclusion of Natural gas under GST. Taxation of CBG injected in CNG and PNG - Exemption of VAT from State Governments required.

4

Integration of LNG terminals, trunk pipelines and CGD areas

5

Easing regulatory compliances such as Change in Land Use and speeding up clearance such as PESO clearance. Also, CGD issues in states (E.g., Pipeline Laying charges)

6

Increasing domestic production of crude oil and natural gas to reduce import dependence

Thank You